A kinetic-impact deflection model

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Learning target: Estimate a velocity change and explain the value of lead time.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 8.*
0 m/s; equal opposite momenta cancel, and the combined mass is 6 kg.

02. Worked example: explain the model
Initial momentum 60 kg·m/s; joined mass 12 kg moves at 5 m/s.

03. Read and interpret the model
For the same added velocity, a longer drift gives a larger sideways offset: $\Delta x = \Delta v \cdot t$. The velocity change is produced by the brief push; it does not require a force throughout the drift.

04. Apply the physics
1 m and 10 m.

05. Explain your reasoning
Ejected material carries momentum, target properties matter and orbital geometry changes the response. Simplified model omits these.

06. Review: connect an earlier idea *Review: Unit 2, Day 15.*
Need orbital geometry/positions and phase or time reference. A period alone does not show simultaneous nearby positions; uncertainties matter.

07. Exit check
Twice the shift in straight-line model; ignores gravity/orbital geometry and target uncertainties.

08. STRETCH: test the model
60/22≈2.73 m/s; same total momentum spread over greater final mass.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 8.*
v_f=5 m/s. Initial K=375 J; final K=37.5 J; fraction 0.10 or 10%.

02. Worked example: explain the model
Initial momentum 60 kg·m/s; joined mass 12 kg moves at 5 m/s.

03. Read and interpret the model
For the same added velocity, a longer drift gives a larger sideways offset: $\Delta x = \Delta v \cdot t$. The velocity change is produced by the brief push; it does not require a force throughout the drift.

04. Apply the physics
Momentum 3.6×10^{6} kg·m/s;Δv 7.2×10^{-4} m/s; shift 720 m. Values are a classroom model, not the measured DART result.

05. Explain your reasoning
Ejected material carries momentum, target properties matter and orbital geometry changes the response. Simplified model omits these.

06. Review: connect an earlier idea *Review: Unit 2, Day 15.*
Overlap days 18-20. Timing is compatible, but spatial geometry, miss distance and uncertainties are still needed.

07. Exit check
Δv=0.005 m/s; impulse 5×10^{6} N·s under the simplified transverse model.

08. STRETCH: test the model
20 N·s; 200 N; subsequent offset 2000 m. This simplified drift ignores gravity and orbital geometry.